GINHAWA · AIR-QUALITY STATION

Station Build Manual: Diagrams

All 15 figures from the build manual, one per page. Follow the full manual for the steps and checks.

1. Block diagram of the station power and data paths (Stage 0: Before you start)
2. How to use a multimeter for voltage and continuity (Stage 0: Before you start)
3. Good and bad solder joints (Stage 0: Before you start)
4. Battery holder with three cells in parallel (Stage 3: Build the battery pack)
5. Wiring of holder, BMS, fuse and INA219 (Stage 3: Build the battery pack)
6. Power board layout with VBAT and GND bus wires (Stage 4: Make the power board)
7. Solar panel and CN3791 wiring (Stage 6: Solar panel and charger (CN3791))
8. TPS63020 and MT3608 wiring and adjustment (Stage 7: 3.3 V converter (TPS63020))
9. ESP32 DevKit V1 pinout with the used pins highlighted (Stage 9: Make the ESP32 board)
10. ESP32 board layout with rails and pull-up resistors (Stage 9: Make the ESP32 board)
11. SEN55 connector pins and their connections (Stage 11: Connect the SEN55 and INA219, then test)
12. A7670SA breakout wiring with antennas and capacitors (Stage 12: Connect the 4G modem (A7670SA) and test)
13. Electrolytic capacitor polarity and parallel connection (Stage 12: Connect the 4G modem (A7670SA) and test)
14. Enclosure drilling plan and internal layout (Stage 14: Prepare the enclosure)
15. Pole mounting with tilted solar panel above the station box (Stage 16: Mount it outside)

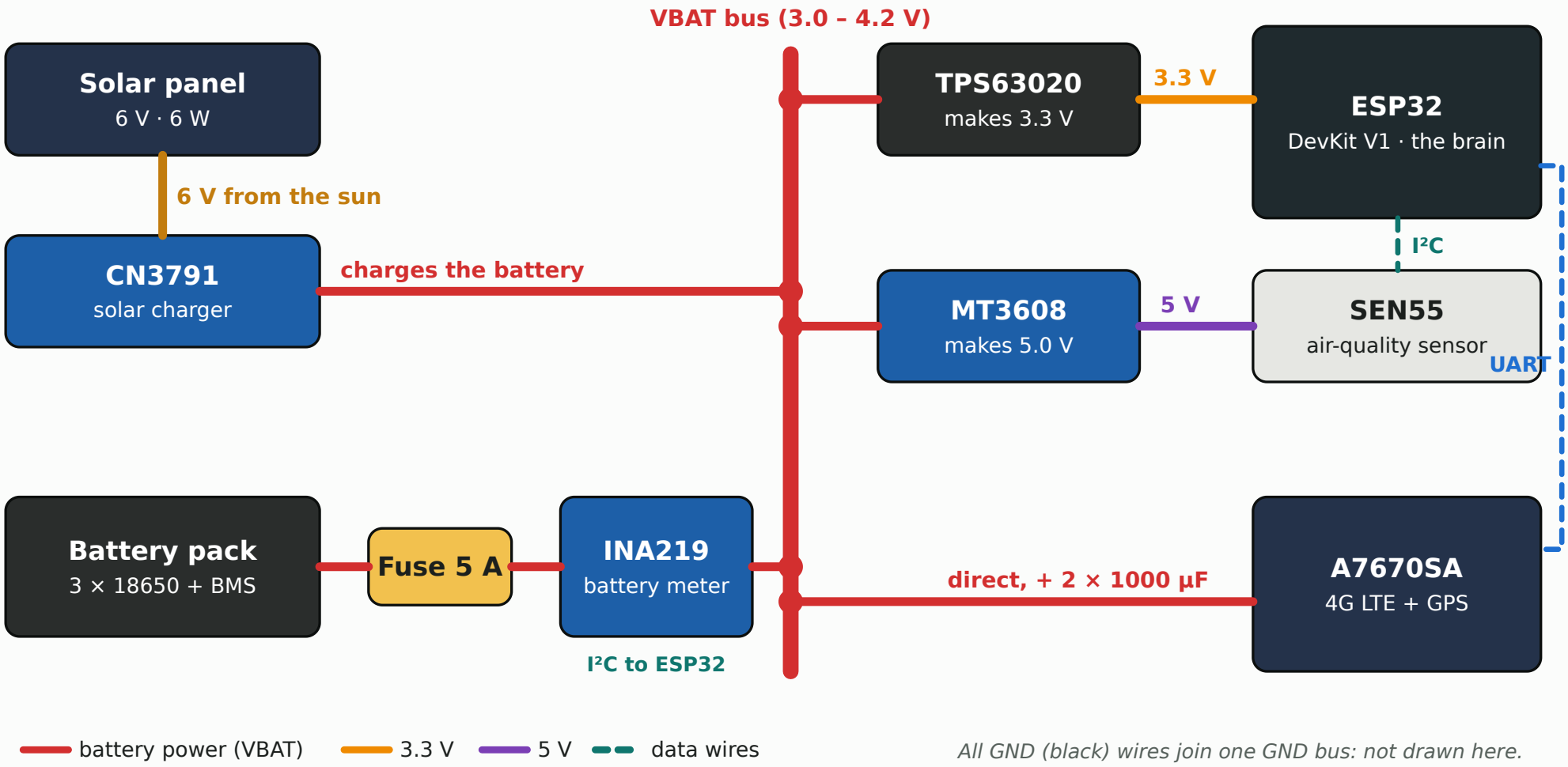


Figure 1. How power and data move through the station. The battery feeds one **VBAT bus**; every module takes its power from that bus. The solar panel refills the battery through the CN3791.

STAGE 0: BEFORE YOU START

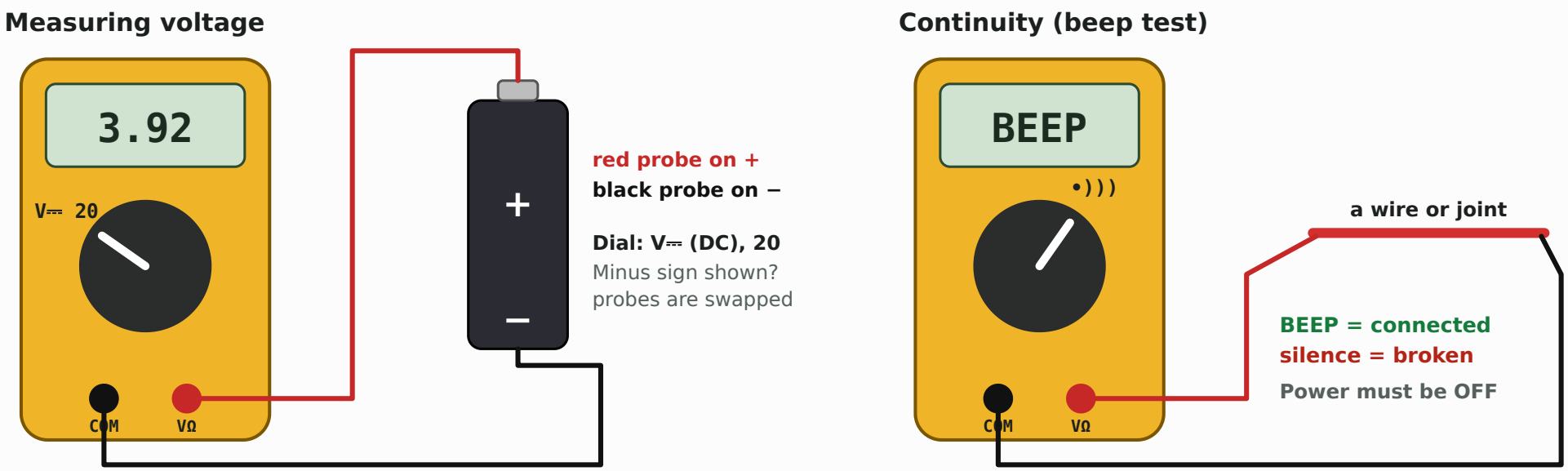


Figure 2. The two multimeter settings you need. Black probe always goes in **COM**, red in **VΩ**.

STAGE 0: BEFORE YOU START

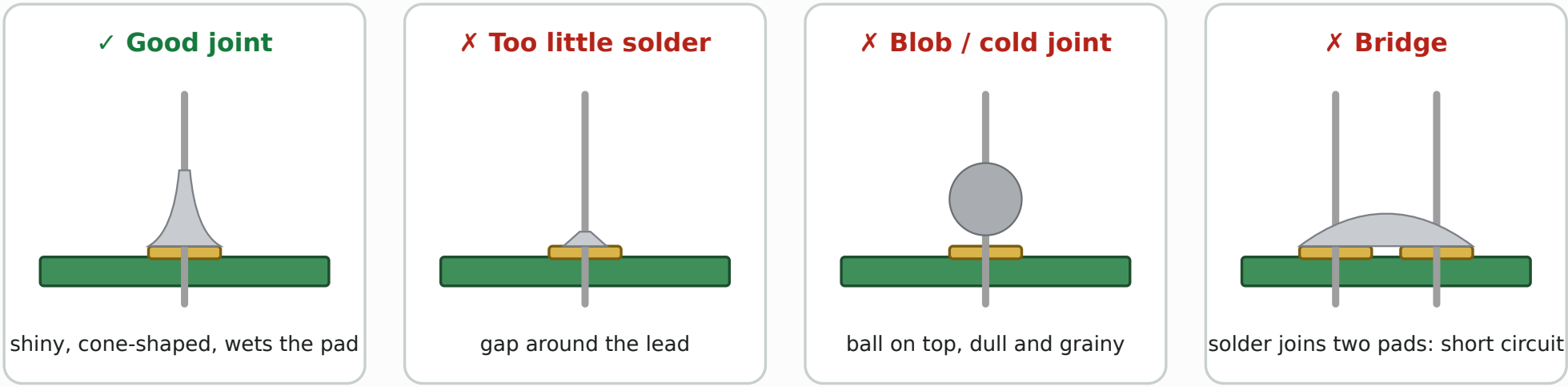


Figure 3. What solder joints should look like. Fix any ✗ joint: reheat it and add or remove solder (use the tip to drag extra away).

STAGE 3: BUILD THE BATTERY PACK

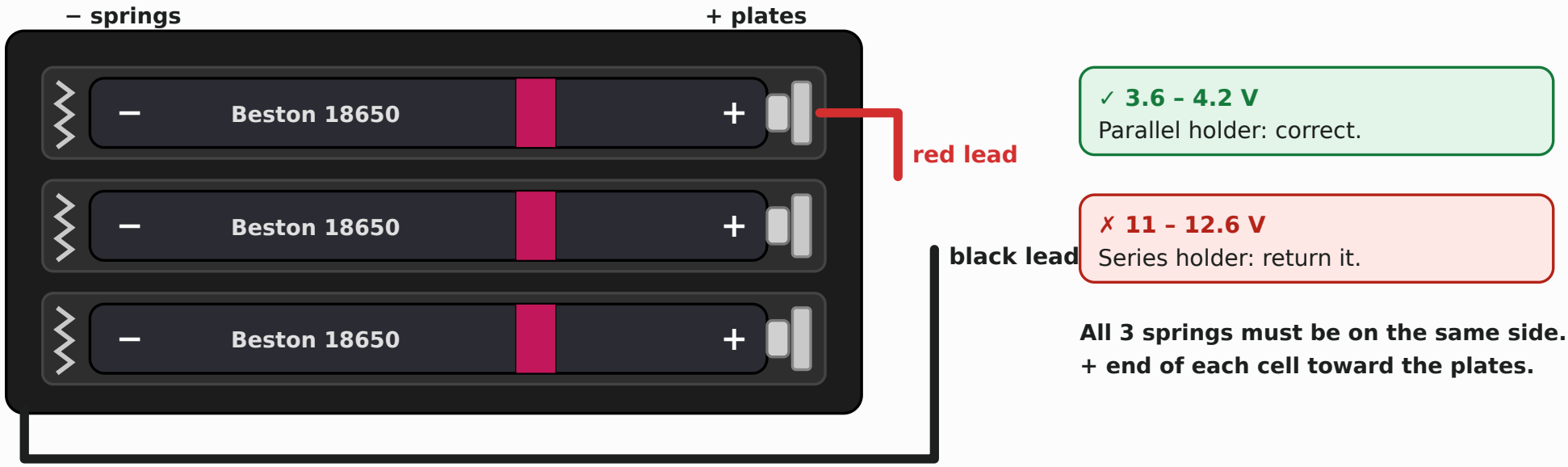


Figure 4. Cells in the 3-cell **parallel** holder. Every cell faces the same way: + end on the metal plates, − end on the springs.

STAGE 3: BUILD THE BATTERY PACK

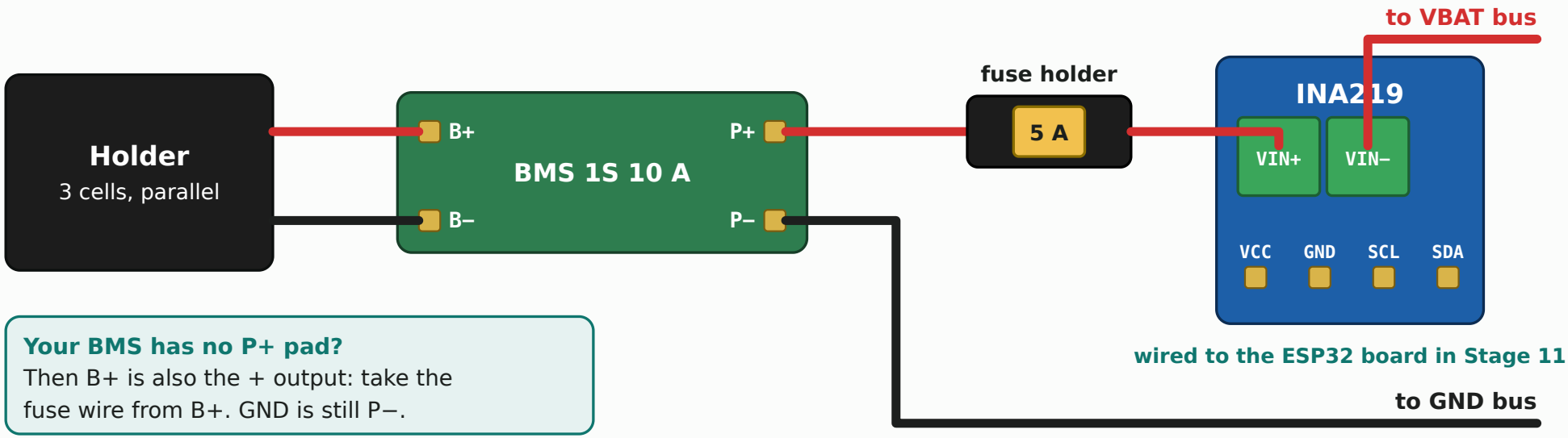
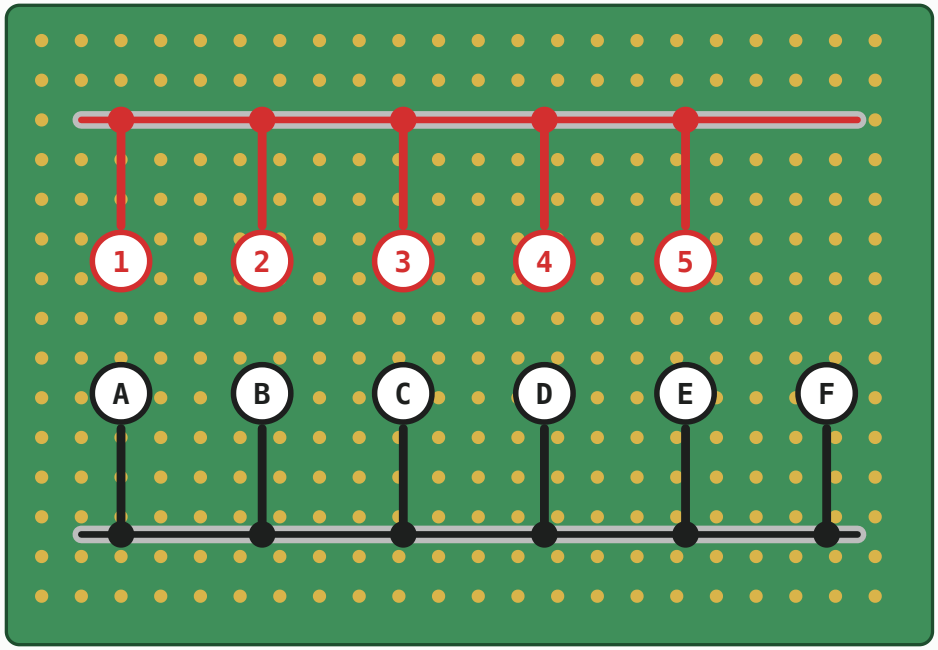


Figure 5. Battery → BMS → fuse → INA219. The thick red path carries all the station's current. **P−** becomes the station's ground (GND).

STAGE 4: MAKE THE POWER BOARD

Red bus (VBAT)
1 INA219 VIN− · 18 AWG
2 CN3791 BAT +
3 TPS63020 VIN
4 MT3608 VIN+
5 A7670SA VBAT · 18 AWG



Black bus (GND)
A BMS P− · 18 AWG
B CN3791 BAT −
C TPS63020 GND
D MT3608 VIN−
E A7670SA GND · 18 AWG
F ESP32 board · 18 AWG

Power board (5 × 7 cm perfboard), seen from the top

Red bus = VBAT, black bus = GND: bare, tinned 18 AWG wire. Taps at least 3 holes apart.

Figure 6. The power board is two long bus wires. Everything that needs battery power solders to the red bus (taps 1–5); every ground solders to the black bus (taps A–F).

STAGE 6: SOLAR PANEL AND CHARGER (CN3791)

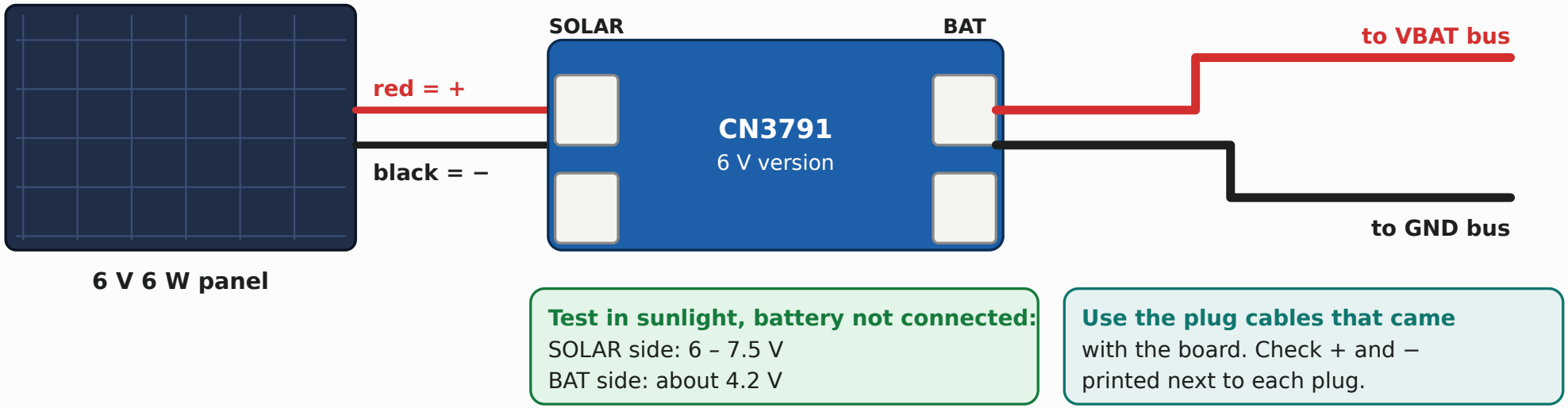


Figure 7. Solar panel → CN3791 → battery bus. Either SOLAR plug works (they are connected together), and the same for BAT.

STAGE 7: 3.3 V CONVERTER (TPS63020)

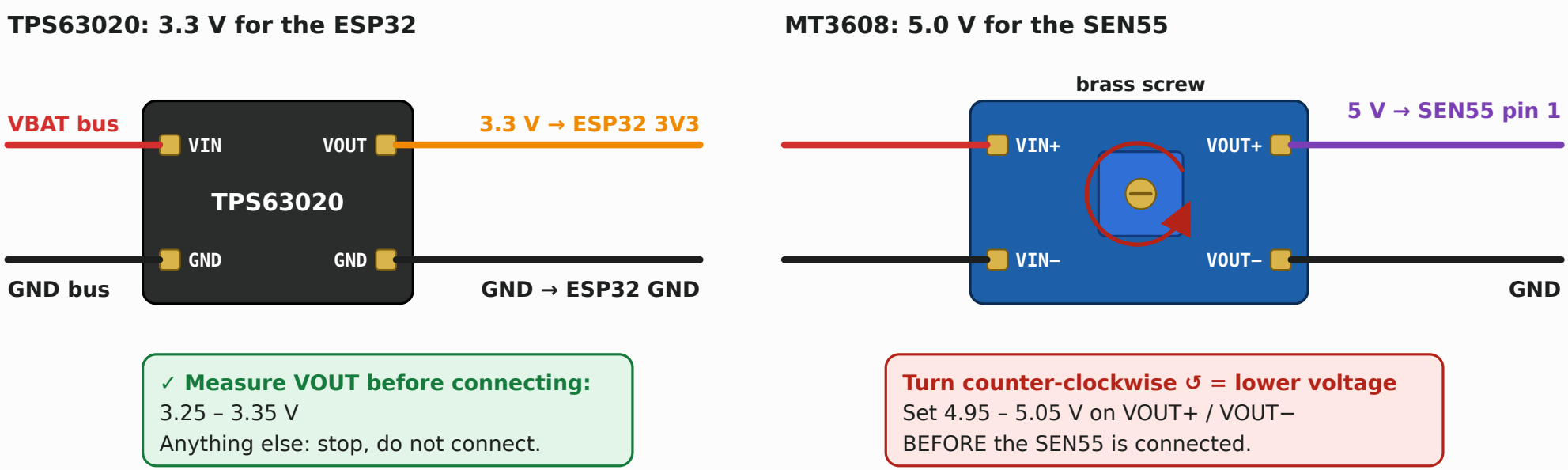


Figure 8. The two voltage converters. Both take power from the VBAT bus. Measure their outputs before anything is connected to them.

STAGE 9: MAKE THE ESP32 BOARD

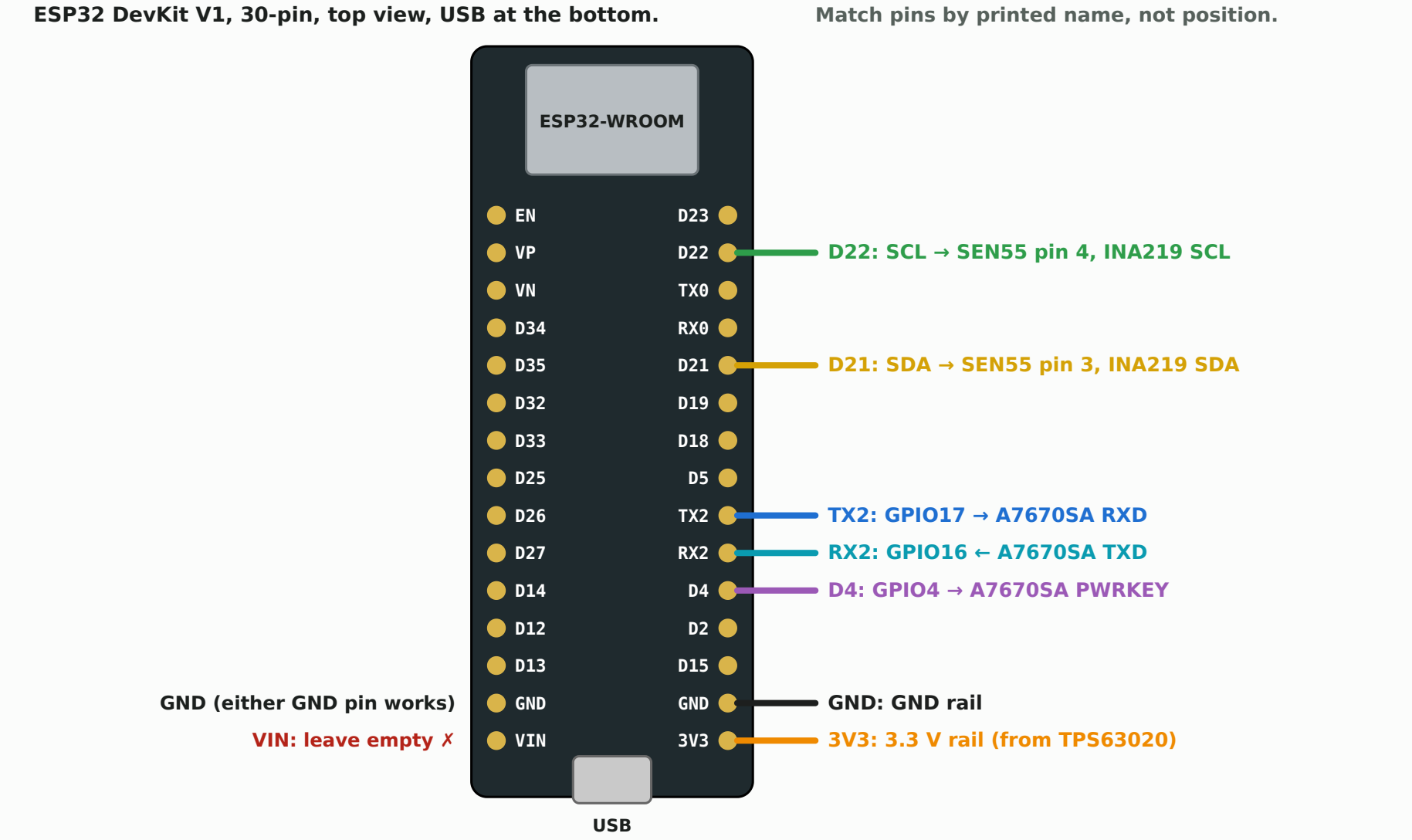


Figure 9. The ESP32 pins this project uses, all on one side of a 30-pin DevKit V1. **TX2 = GPIO17** and **RX2 = GPIO16** on the board's labels.

STAGE 9: MAKE THE ESP32 BOARD

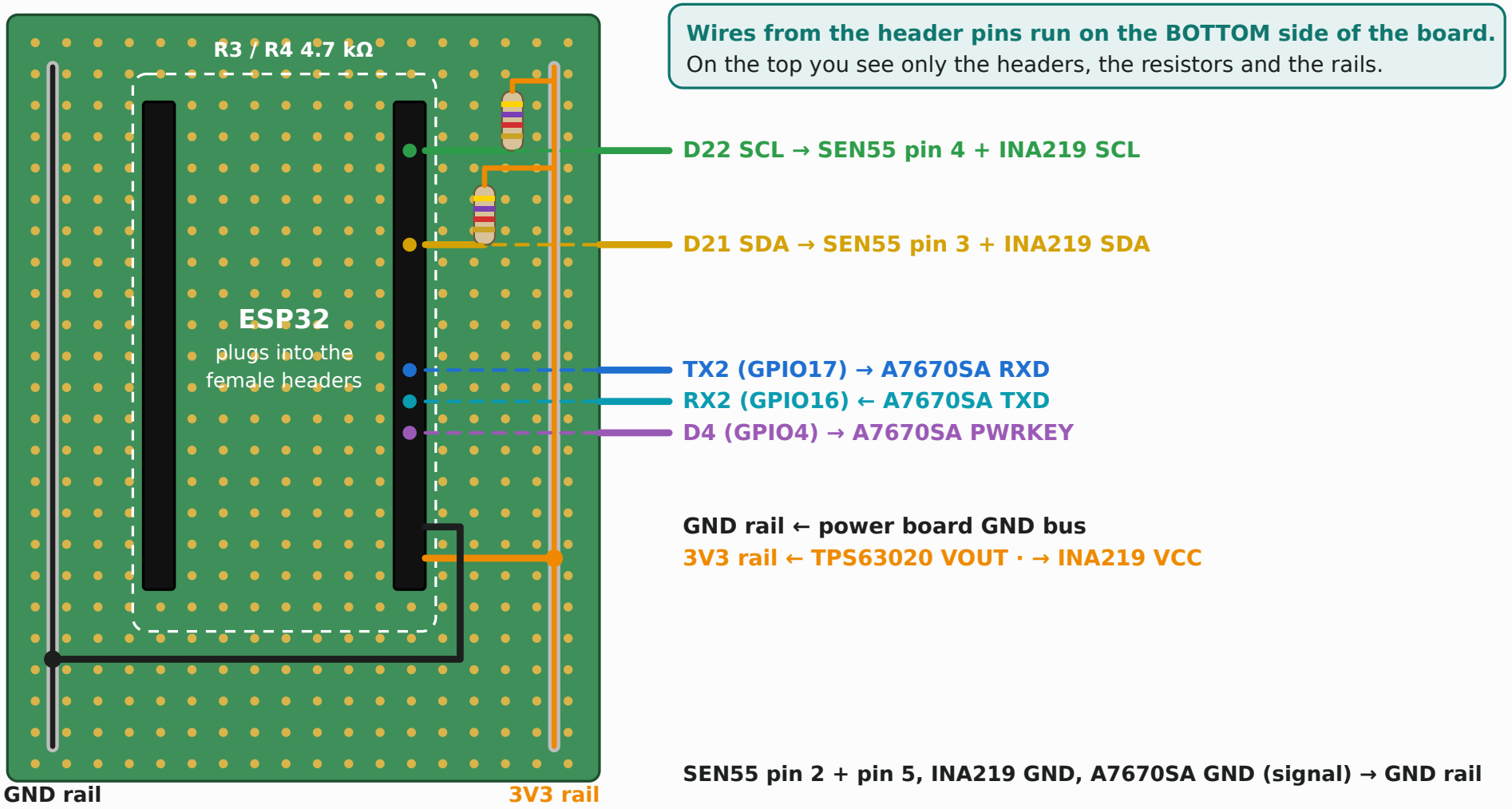


Figure 10. The ESP32 board (7 × 9 cm). Two rails (GND and 3V3), two pull-up resistors, and solder points for the sensor and modem wires.

STAGE 11: CONNECT THE SEN55 AND INA219, THEN TEST

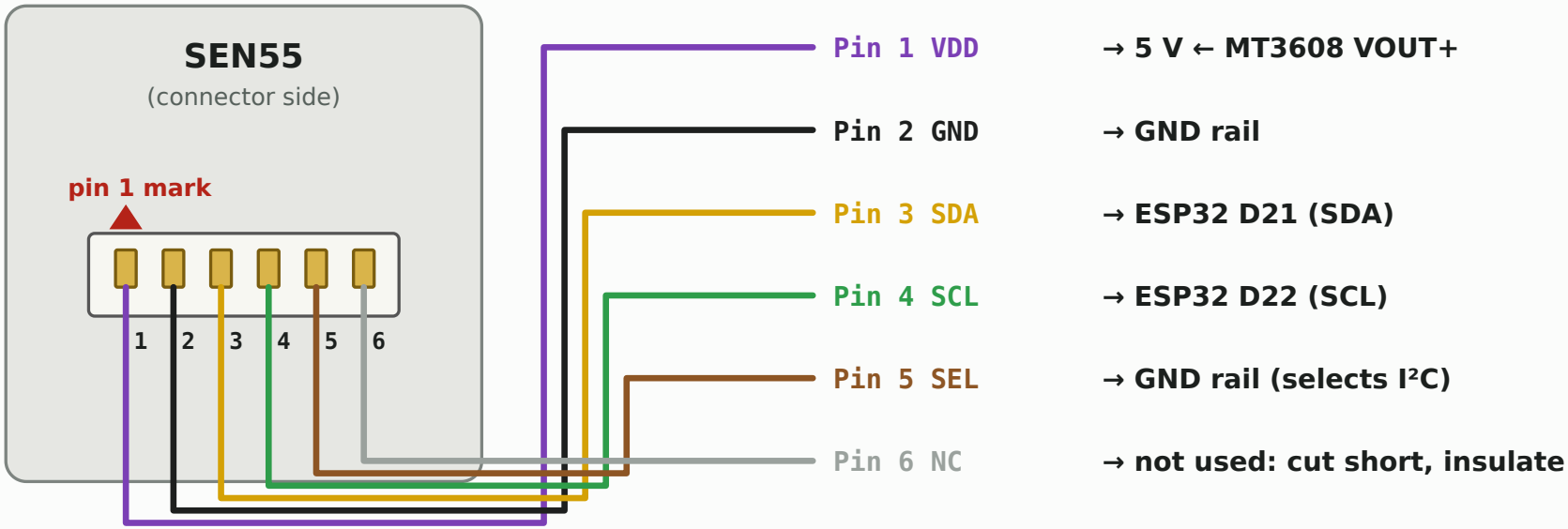


Figure 11. SEN55 pin-out and where each wire goes. Find **pin 1** from the mark printed next to the sensor's connector, then count. Go by pin number, never by wire colour.

STAGE 12: CONNECT THE 4G MODEM (A7670SA) AND TEST

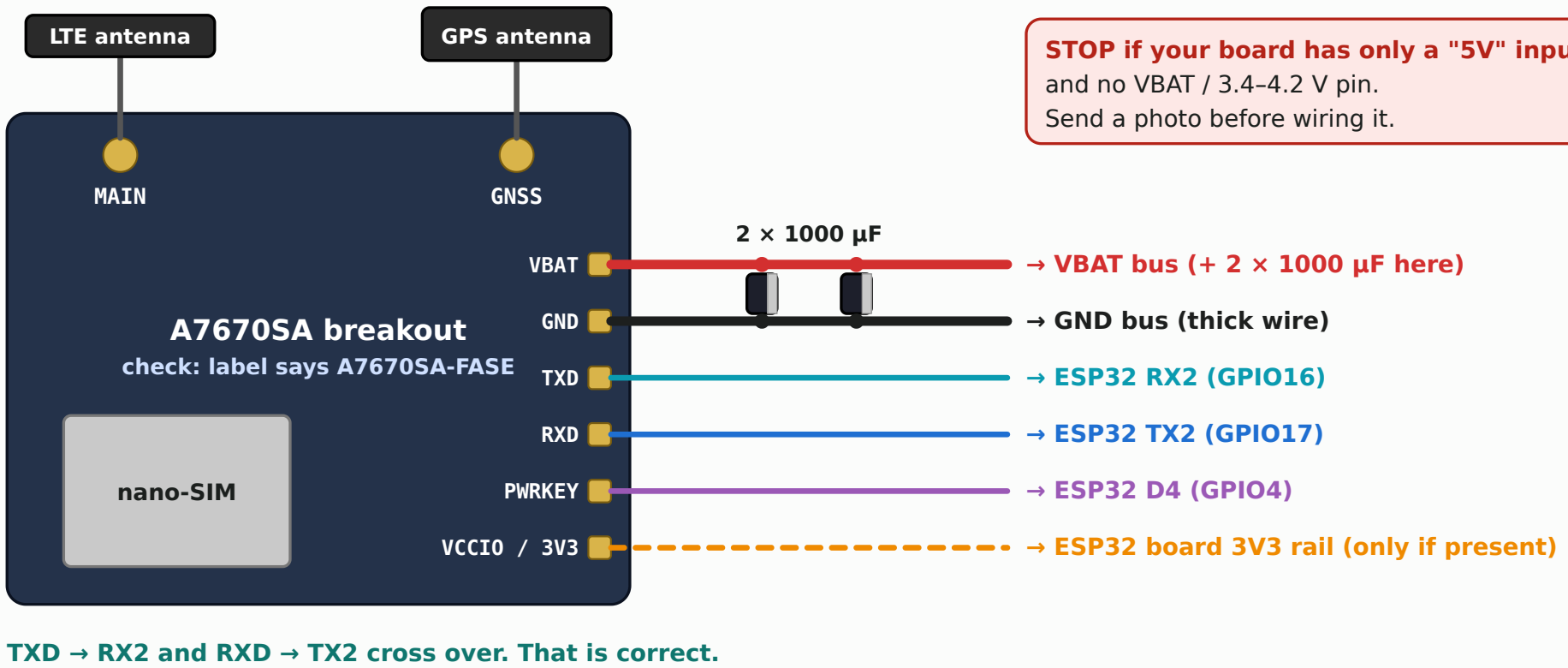


Figure 12. A7670SA connections. Pin names differ between breakout makers: PWRKEY may be printed as **PWR**, **PEN** or **K**; VBAT may be **VCC 3.4–4.2 V** or **BAT**.

STAGE 12: CONNECT THE 4G MODEM (A7670SA) AND TEST

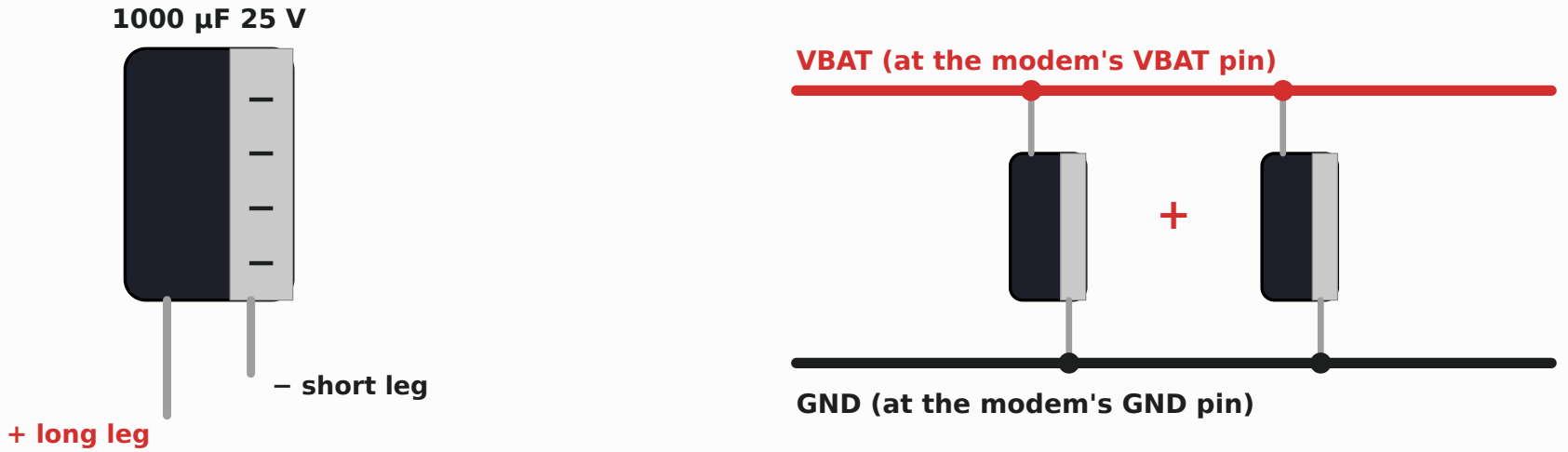
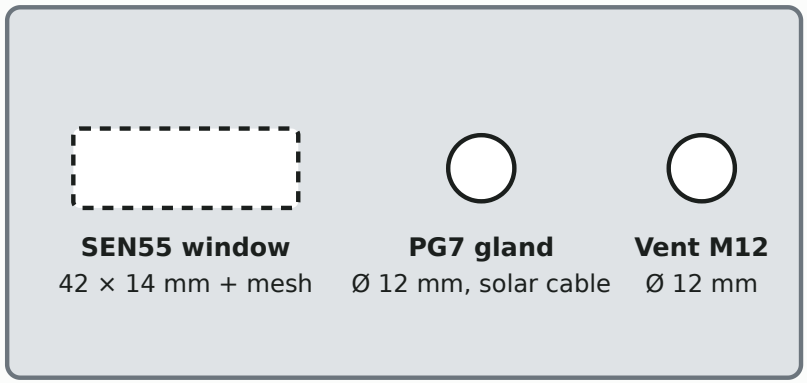


Figure 13. Capacitor polarity. The side with the stripe is **−**. Two capacitors side by side, both + to VBAT and both − to GND.

STAGE 14: PREPARE THE ENCLOSURE

Bottom face (150 × 70 mm box shown)



Drill on the bottom (or low on the sides).
Never drill the top: rain sits there.

ABS / plastic boxes let radio signals through.
Flat or patch antennas can stay inside the box.
Only the solar cable needs a gland.

Inside, lid off, box standing upright

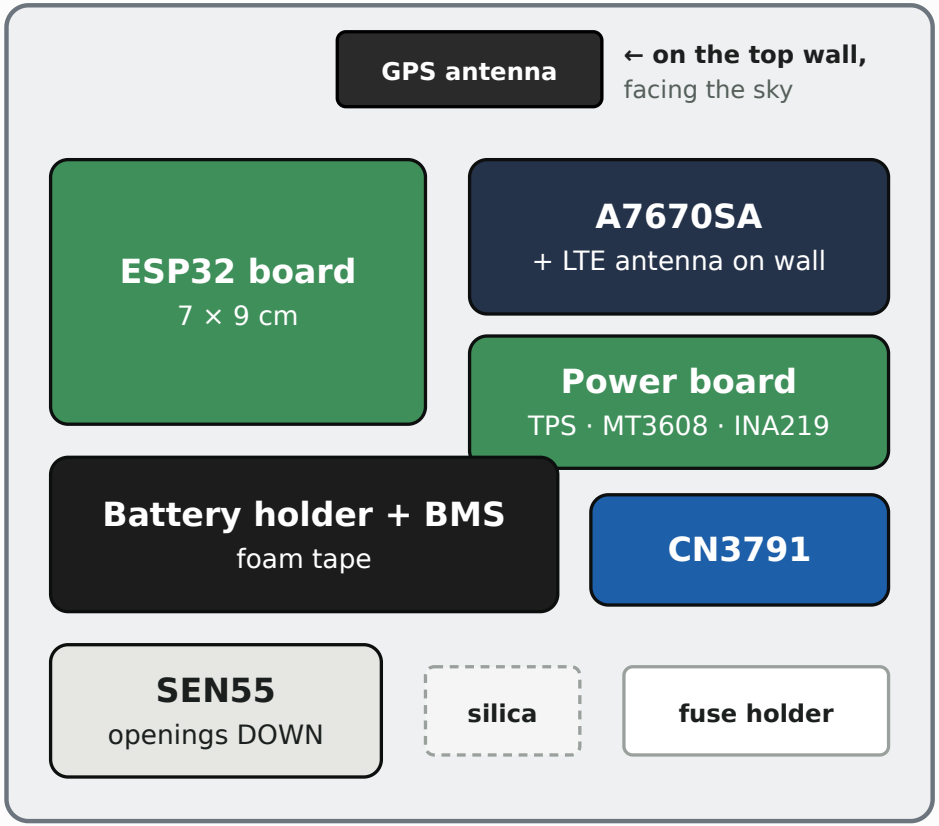


Figure 14. Drilling plan and layout inside an IP65 box of about 150 × 100 × 70 mm. Adjust the positions to the box you buy, keeping these rules.

STAGE 16: MOUNT IT OUTSIDE

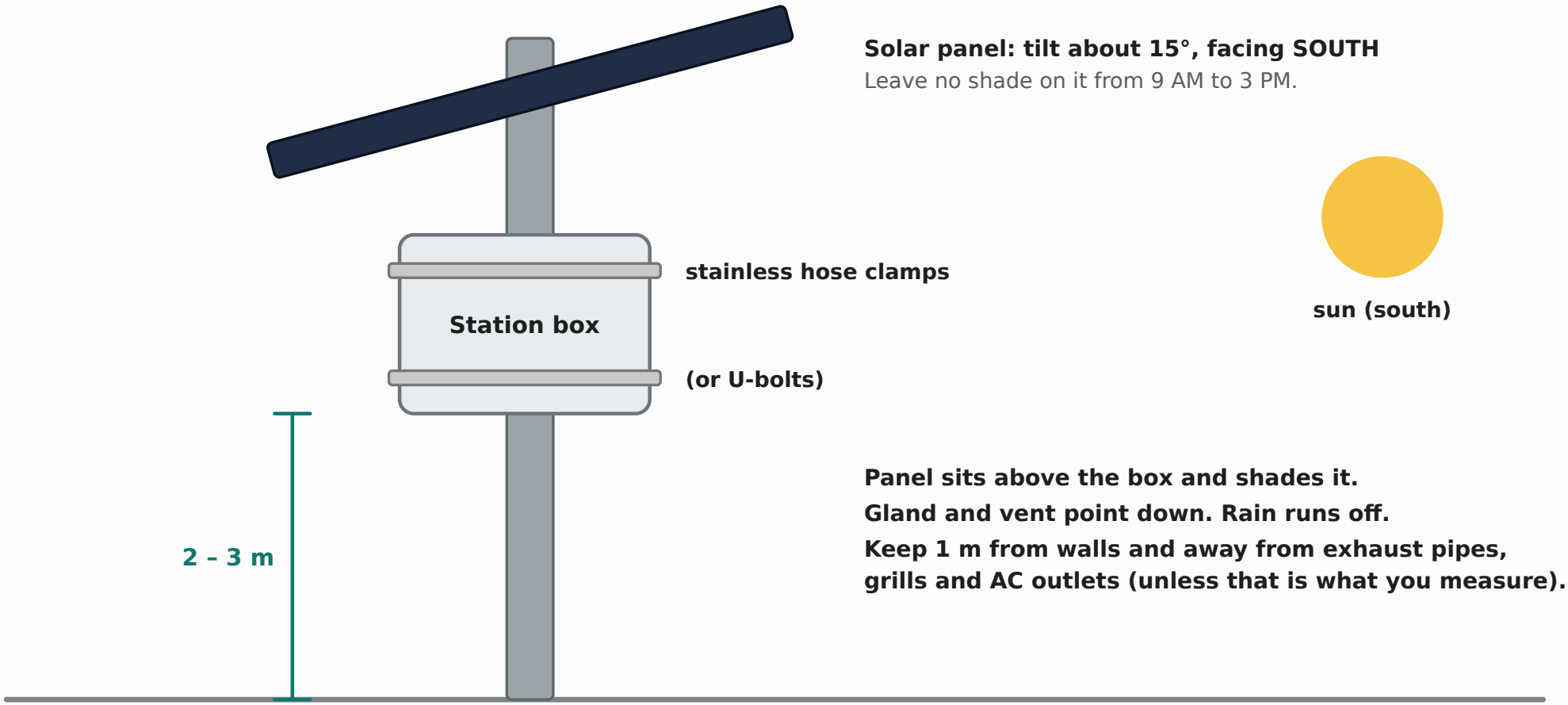


Figure 15. Mounting on a pole. The sensor should breathe the same air people do: 2 to 3 m above the ground, in open air.